

OMKAR WAGHOLIKAR

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EDUCATION

Pune Institute of Computer Technology

Bachelor of Engineering - Computer Engineering

July 2021 - July 2025

CGPA: 8.8

TECHNICAL SKILLS

Languages: Java, Python, JS, C#, TypeScript

Databases: PostgreSQL, MongoDB, MySQL, Redis

Frameworks: Django, DRF, React, Node.js, Celery

Data Engineering: Airflow, DBT, BigQuery, GCS

Technologies: Docker, Kafka, RabbitMQ, Elastic Search

AI Tools: Claude Code, GitHub Copilot, AWS Bedrock

EXPERIENCE

Associate Software Development Engineer, NiCE Ltd

July 2025 – Current

- Optimized core backend application logic in **C#**, improving **performance and reliability** of critical paths.
- Fixed infinite reprocessing issue in **DLQ** by debugging and correcting **routing logic** to prevent retries.
- Designed document ingestion pipelines with checkpointing, batch processing, retry logic, and failure handling.
- Leveraged AWS services like **Bedrock, Lambda, Batch** for building **scalable, event-driven** cloud solutions.

Project Assistant, Pune Knowledge Cluster

October 2024 – February 2025

- Architected and deployed a full-stack platform** for a **national-level** data challenge competition.
- Built the frontend in **React** and backend in **Django REST Framework**, with secure **token-based auth**.
- Designed a **scalable submission pipeline** handling large user file uploads via **Azure object storage**.
- Created internal tools for communication, contact registration, and **workflow automation**.

Software Engineering Intern, Digital and Technology, General Mills India

June 2024 – August 2024

- Worked with the **Data Engineering** team on a commercial **File Data Ingestion** project utilizing BigQuery.
- Established standardized procedures for incoming file management via shared mailbox and **GCP Buckets**.
- Executed the FDI process with **data quality checks** using **DBT**, integrated into **Airflow DAGs**.

Research Associate, Pune Knowledge Cluster

November 2023 – February 2024

- Developed solutions for Central Government Problem Statement using **Docker, Elastic Search**, and **DRF**.
- Led the **full project lifecycle** from requirement analysis to system design, development, and deployment.
- Implemented a custom **Information Retrieval System** using **RAG** to assist social healthcare workers.

PROJECTS

EduNexus: Academic File Sharing Platform (Django, React, TypeScript, Celery, Firebase, pgvector)

- Built a **full-stack platform**: React SPA (Tailwind, React Query, drag-and-drop uploads, inline PDF.js previews) over a Django REST API on Postgres + pgvector, with Firebase auth and signed download URLs.
- Implemented **hybrid lexical, semantic search** over PDF/PPTX/DOCX, combining **BM25** with **pgvector sentence-transformer embeddings**; surfaced via debounced autocomplete, and per page anchors.
- Engineered an **ingestion pipeline** with **OCR, LLM summaries, hash based deduplication**
- Hardened for production: **Firebase Admin SDK token verification, role-based access control**.

Project MAVIS: Media Authenticity Verification and Integrity System (Django, OpenCV, PyTorch)

BE Project

- Establish **provenance and media authenticity** using system Leveraged **hardware-backed security**, advanced **steganographic and cryptographic** technologies.
- Engineered multi layered approaches using Wavelet DCT, GAN based steganographic watermarking and Reed Solomon encoding, ensuring tamper resistance and robust data embedding against various distortions.

BRAGS: Build your own RAGs (Python, Go, LangChain)

- Python package** for easily deploying custom **Retrieval-Augmented Generation (RAG)** pipelines.
- Integrated a background **Go file watcher** to continuously monitor a document folder and keep the **vector database** synchronized in **real time**.
- Combined **Python for RAG orchestration** with a **Go watcher process** to automate **document ingestion** and **vector index updates** by listening and debouncing **OS level events**.

EXTRA-CURRICULAR ACTIVITIES

- Hackathons**: 2 x Winner at PICT InC' 2025, Nexus 2.0 (AIT) | Top 5 at Innovation '25 (Cummins COE).
- Open Source Contributor**: **CircuitVerse**: 3 merged PRs (bug fixes, refactoring) and 1 issue reported.
- Accepted speaker for a talk on **Unix file descriptors and PTY internals** at **FOSS United Pune**, April 2026.

CERTIFICATIONS

Deep Learning Specialization | Basic GANs | GCP Core Infrastructure | LFD103: Linux Kernel Development